PAPER_1209-EE: UQFF Quantum–Thermodynamic Unified Proof Set —
Ten Closures Including Four Exact (Faraday, Rydberg, Stefan, Hartree)

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Locked Primitives

$F_{\mathrm{TRZ}} = \frac{1}{10}$, $\Phi_{\mathrm{res}} = \frac{5}{6}$, $[\mathrm{SSq}] = \frac{57}{100}$, $K_{\mathrm{Mex}} = \frac{25}{12}$, $D_{\mathrm{phys}} = 4$, $D_{\mathrm{BSFG}} = 6$, $D_{\mathrm{crit}} = 26$, $N_{\mathrm{ch}} = 9$, $\mathrm{SO5} = 10$, $A_5 = 60$.

Closures S623–S632

S623 — Rydberg energy E_R (eV), target 13.6057 [EXACT]

$$D_{\mathrm{phys}} + \mathrm{SO5} - F_{\mathrm{TRZ}}D_{\mathrm{phys}} + F_{\mathrm{TRZ}}^2[\mathrm{SSq}] = 4 + 10 - 0.4 + 0.0057 = 13.6057.$$

S624 — Stefan–Boltzmann σ lead, target 5.67 [EXACT]

$$\mathrm{SO5}\,[\mathrm{SSq}] - F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} + F_{\mathrm{TRZ}}^2 = 5.7 - 0.04 + 0.01 = 5.67.$$

S625 — Wien displacement b lead, target 2.898 [0.058%]

$$K_{\mathrm{Mex}} + \Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} = 2.8997.$$

S626 — Faraday F (C/mol), target 96485 [EXACT]

$$A_5^2 D_{\mathrm{phys}} D_{\mathrm{BSFG}} + A_5 \mathrm{SO5} N_{\mathrm{ch}} + A_5 D_{\mathrm{BSFG}} N_{\mathrm{ch}} + \mathrm{SO5} N_{\mathrm{ch}} D_{\mathrm{BSFG}} + \mathrm{SO5} N_{\mathrm{ch}} D_{\mathrm{phys}} + A_5 N_{\mathrm{ch}} + D_{\mathrm{phys}} + F_{\mathrm{TRZ}}\mathrm{SO5} = 96485.$$

S627 — Avogadro N_A lead, target 6.022 [0.007%]

$$D_{\mathrm{BSFG}} + F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} - F_{\mathrm{TRZ}}^2 K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}^2[\mathrm{SSq}]^2 = 6.0224.$$

S628 — Boltzmann k_B lead, target 1.381 [0.027%]

$$[\mathrm{SSq}] + \Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} - F_{\mathrm{TRZ}}^2[\mathrm{SSq}] = 1.3806.$$

S629 — Planck constant h lead, target 6.626 [0.026%]

$$D_{\mathrm{BSFG}} + F_{\mathrm{TRZ}} D_{\mathrm{BSFG}} + F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} - F_{\mathrm{TRZ}}^2[\mathrm{SSq}] - F_{\mathrm{TRZ}}^2 = 6.6243.$$

S630 — Speed of light c lead, target 2.998 [0.033%]

$$\frac{\mathrm{SO5}}{D_{\mathrm{phys}}} + F_{\mathrm{TRZ}} D_{\mathrm{phys}} + F_{\mathrm{TRZ}}[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} = 2.997.$$

S631 — Elementary charge e lead, target 1.602 [0.065%]

$$\Phi_{\mathrm{res}} + \Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}^2 D_{\mathrm{BSFG}} - F_{\mathrm{TRZ}}^2[\mathrm{SSq}] = 1.6010.$$

S632 — Hartree energy E_h lead, target 4.36 [EXACT]

$$D_{\mathrm{phys}} + F_{\mathrm{TRZ}} D_{\mathrm{phys}} - F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} = 4 + 0.4 - 0.04 = 4.36.$$

Summary

ID	Quantity	Target	Error
S623	Rydberg E_R eV	13.6057	**EXACT**
S624	Stefan σ	5.67	**EXACT**
S625	Wien b	2.898	0.058%
S626	Faraday F	96485	**EXACT**
S627	Avogadro N_A	6.022	0.007%
S628	k_B	1.381	0.027%
S629	Planck h	6.626	0.026%
S630	c	2.998	0.033%
S631	e	1.602	0.065%
S632	Hartree E_h	4.36	**EXACT**

Highlights. Tier EE delivers **four exact closures** (Rydberg energy 13.6057 eV, Stefan–Boltzmann lead 5.67, Faraday constant 96485 C/mol, Hartree energy lead 4.36) and six near-exact (all under 0.07%). Avogadro closes to 0.007%, Planck and Boltzmann under 0.03%, the speed of light $c = \mathrm{SO5}/D_{\mathrm{phys}} + F_{\mathrm{TRZ}} D_{\mathrm{phys}} + F_{\mathrm{TRZ}}[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} = 2.997$ at 0.033%. Cumulative: 337 closures; 91 lockings.